Bell Inequality in Quantum Magnets

THE OHIO STATE UNIVERSITY

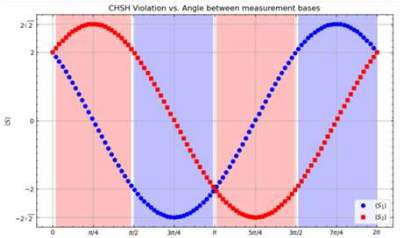
Evan Waddell, Ryan Buechele

CHSH Inequality

The **CHSH** inequality is a type of Bell inequality. We have 2 people, Bob and Alice. There is a pair of entangled qubits, one is sent to Alice, the other to Bob. Alice and Bob both randomly choose to measure their qubit in either the x or z direction giving results of +1 or -1, and we add take the average of these measurements. Classically, the results should follow the formula

$$|\langle A_x B_x\rangle + \langle A_x B_y\rangle + \langle A_y B_x\rangle - \langle A_y B_y\rangle| \leq 2.$$

, but when performing this experiment, it is easily shown that the values can exceed 2 and be as high as $2\sqrt{2}$. This has been proven many times in laboratory settings, but what does this mean?

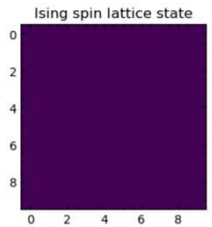


Local Realism

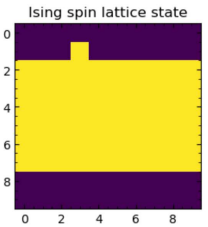
- Locality- Information cannot travel faster than the speed of light.
- Realism- Objects have definite values without being measured.
- **CHSH** inequality demonstrates limits on theories with local realism;
  The violation of this inequality proves one or both components not to be true.

Classical Ising Ferromagnets

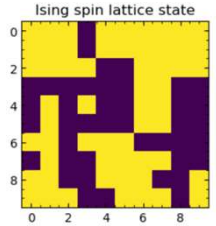
$$\mathcal{H} = -J\sum_{\langle j,k\rangle} s_j s_k - h\sum_j s_j. \quad (s_j = \pm 1)$$



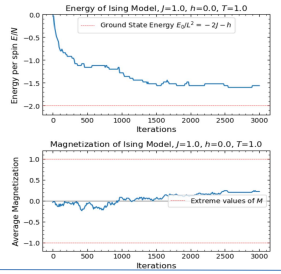
Low Temp (0.1)

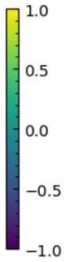


Med Temp (1)



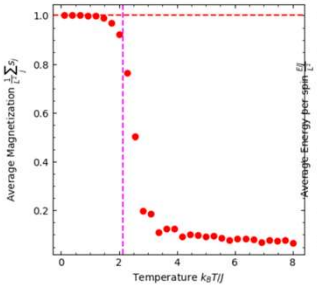
High Temp (3)

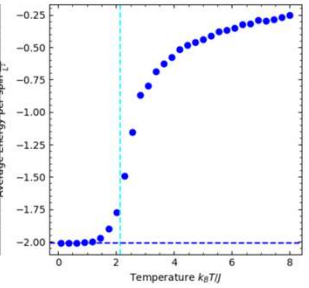




We used the **Metropolis Monte Carlo algorithm** to find the ground state of our system at different temperatures. In general, it should look as such:

- Low Temp- Spins align
- High Temp- Spins disorganized
- Temps Close to Transition-
  Spins in large aligned regions

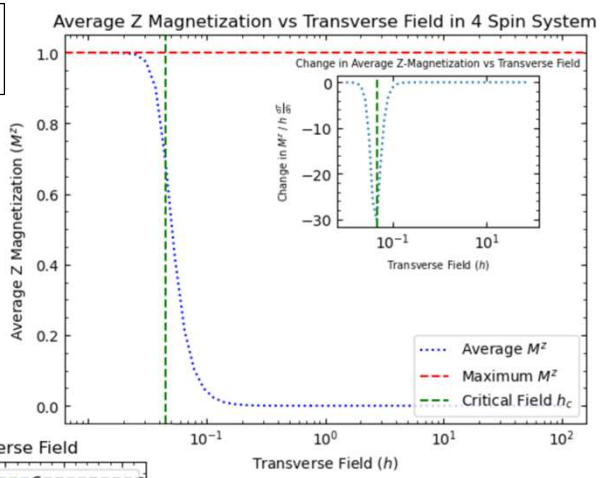


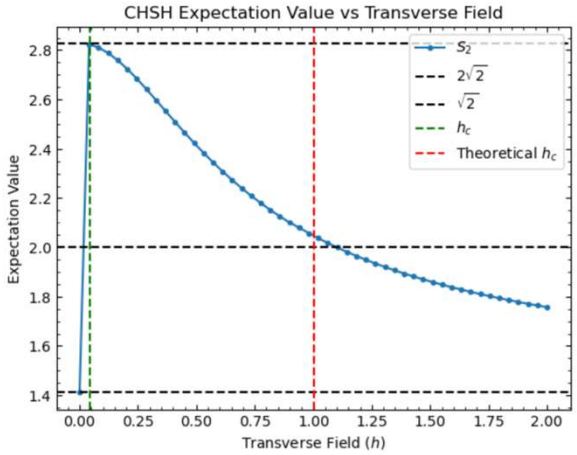


Quantum Ising Ferromagnets

$$\mathcal{H} = -J\sum_{\langle j,k\rangle} \sigma_j^z \sigma_k^z - h\sum_j \sigma_j^x.$$

The first thing we find is the phase transition in our quantum system. We achieve this by finding where the change in the average Z magnetization in the field is the greatest. In this model, it was found to be where $h = 0.045$. This is where the ferromagnet transitions into a disordered paramagnet.





This is the culmination of everything mentioned before. We now use the ground state of the system as our starting state and view the measurements as we change the angle and the magnetic field of the system. The critical field in our simulation was affected by small system size, but the **CHSH** violation happens very close to the theoretical transition in an infinite system $h = 1$

Bibliography

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CHSH: Clauser, J. F., Horne, M. A., Shimony, A., & Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters, 23(15)*, 880.

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